AMIDI: AUDIO-SUPERVISED EVENT MODELING FOR EXPRESSIVE MIDI GENERATION

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ABSTRACT

Symbolic music models produce editable scores, but their training targets often omit continuous performance cues. Audio models retain these cues, yet their outputs are hard to edit note by note. We present AMIDI, a controlled study of audio-supervised pretraining for expressive MIDI generation. Both systems share a causal Transformer and an expressive event vocabulary for notes, velocity, duration, instrumentation, CC1, and sustain pedal. Pretraining audio is first transcribed to pseudo-MIDI for both systems; the experimental system additionally predicts frozen EnCodec tokens at 50 Hz from causal MIDI states and performance state. The audio objective therefore provides an additional training signal to the shared MIDI representation. Both systems are then trained identically on higher-quality paired MIDI–audio data. We describe the protocol and a demo of MIDI-only versus audio-supervised pretraining via generated MIDI, controllers, and codec audio.

1. MOTIVATION

Symbolic music models offer editable outputs but depend on expressive symbolic targets. Prior work finds velocity and pedal tokens lower FAD vs. note-only baselines [1]. Large corpora such as Lakh MIDI provide scale but heterogeneous controller data [2]; MAESTRO provides high-quality performance MIDI but is limited to classical piano [3]. GiantMIDI-Piano expands piano MIDI through AMT, whose outputs still inherit transcription errors and representation limits [4].

Audio representations retain complementary performance information. MERT and JukeMIR show useful musical information in learned audio representations, while UniAudio demonstrates large-scale audio-token generation [5–7]. This motivates using audio as auxiliary supervision for editable symbolic generation.

AMIDI emits MIDI with velocity/CC1 and sustain pedal. We ask whether an auxiliary audio-token objective improves an otherwise identical expressive MIDI LM. During pretraining, both branches use the same AMT-derived MIDI targets; only the experimental branch also receives paired 50 Hz audio-code supervision. Both then use the same MIDI+audio training on higher-quality paired alignment data.

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2. SYSTEM DESIGN

2.1 Expressive MIDI event representation

We represent MIDI as an autoregressive sequence of expressive events quantized at 10 ms resolution, with up to 32 independent tracks. A sequence begins with track and program definitions. Time advances through variable-length `SHIFT` tokens, followed by the structured event payloads shown in Fig. 2. Each note event specifies track, pitch, velocity, and duration, while CC1 and sustain-pedal events update the corresponding controller state.

The vocabulary contains 4,558 tokens. Songs are divided into event-aligned chunks of at most 8,192 tokens. Later chunks repeat the track header as masked context, while each next-token target appears in exactly one chunk.

Both systems use a 24-layer, 896-dimensional GPT-2-style decoder with 14 attention heads (242.9M parameters), trained with a causal next-event objective.

2.2 Causal 50 Hz audio supervision

The experimental branch uses a frozen EnCodec tokenizer [8] in the 32 kHz, four-codebook, 50 Hz configuration used by MusicGen [9]. For frame f, the conditioner gathers the last MIDI hidden state whose cumulative event time does not exceed $20f$ ms. It also reconstructs all notes active at f, their track, pitch, velocity, duration, normalized age and remaining duration, plus each track's program, latest CC1, and pedal value.

The conditioner pools active-note and controller vectors into a 512-dimensional frame representation. A 4-layer causal audio decoder consumes this representation and the previous audio codes. A gated 12-layer causal refiner adds capacity. Writing the frame-level MIDI condition as c_f, the residual codebooks are predicted sequentially:

$$p(q_f \mid q_{<f}, c_f) = \prod_{k=1}^{4} p(q_f^k \mid q_{<f}, q_f^{<k}, c_f). \qquad (1)$$

The ground-truth earlier codebooks condition training; sampled codebooks condition inference. No discrete MIDI sample lies on the audio-loss gradient path.

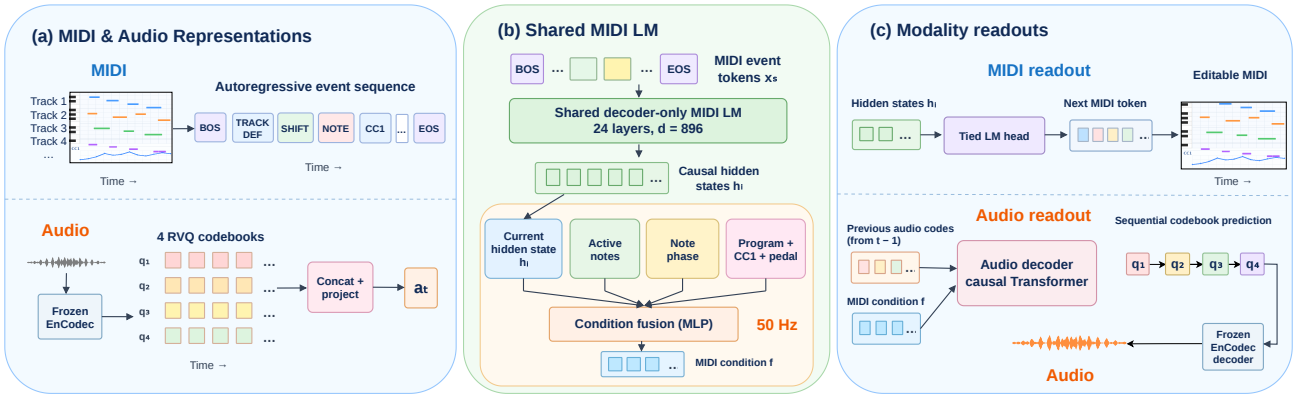


Figure 1. AMIDI event–audio architecture. (a) MIDI events and EnCodec RVQ codes. (b) Shared causal MIDI LM with a 50 Hz conditioner. (c) Tied MIDI readout and causal audio decoder.

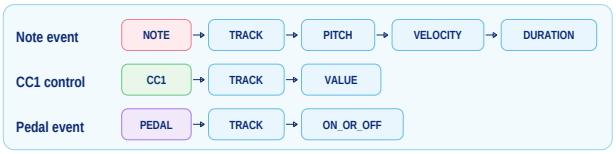


Figure 2. Event payloads. Each note, CC1 update, or pedal change is a short token sequence.

2.3 Objectives

The MIDI objective is unweighted next-token cross-entropy $\mathcal{L}_{\text{MIDI}}$; repeated headers and padding are masked. The audio loss averages cross-entropy over valid frames and all four codebooks. With audio supervision,

$$\mathcal{L} = \mathcal{L}_{\text{MIDI}} + 0.20\mathcal{L}_{\text{audio}}. \tag{2}$$

Separate AdamW optimizers update the MIDI LM and audio modules. During PT, the control uses only $\mathcal{L}_{\text{MIDI}}$ and the experiment uses the summed loss; during alignment, both use the summed loss.

3. DATA AND PROTOCOL

The PT pool combines MusicNet, tagged Jamendo excerpts [10], and public-domain Musopen recordings. We run AMT on all recordings and use the resulting pseudo-MIDI for both systems; MusicNet contributes about 34 hours of aligned classical audio [11]. The event index has 6,950 training and 363 validation songs after song-level splitting. Only the experimental system additionally receives the original audio-token loss during PT.

Both groups then use the same paired MIDI–audio alignment pool. MID-FiLD contributes 3,990 official training MIDIs rendered through a CC1-crossfade SFZ pipeline with native CC1 [12]. URMP contributes 44 real ensemble recordings and 149 stems [13]; stem MIDI is rebuilt from aligned note annotations with pseudo-CC1, and each mix MIDI is formed by merging its aligned stem MIDIs rather than using the non-performance-aligned score MIDI. The classical set has 67 chamber scores (about 4.9 h) with synthetic, slew-limited CC1. Filtered Lakh adds 182 sixty-second windows (about 3.0 h) with the same synthetic CC1 and SFZ pipeline [2]. MAESTRO v3 adds real piano audio with performance MIDI including velocity and sustain [3].

Table 1. Data used in pretraining and alignment.

Stage	Data	Notes
PT	MusicNet, Jamendo, Musopen	AMT MIDI for both; audio loss for experiment
Alignment	MID-FiLD	Rendered audio using dataset CC1
	URMP	Aligned stem MIDI; merged mix MIDI
	music21 classical	Synthetic CC1 + SFZ rendering
	Filtered Lakh	Synthetic CC1 + SFZ rendering
	MAESTRO	Real piano audio + performance MIDI

We match MIDI tokens per epoch and compare checkpoints at similar MIDI validation NLL to control for differences in symbolic-model fit.

4. DEMONSTRATION AND EVALUATION

Both systems are evaluated on continuation from the same prompts. Music-major students perform blind, randomized pairwise preference tests on (i) shared-library renderings of control vs. experimental MIDI and (ii) each model's native codec audio. Since alignment training is identical, the controlled difference is audio supervision during PT. We evaluate generated continuations by human preference rather than similarity to a reference continuation, since multiple musically valid continuations may differ substantially from the reference.

The demo shows matched continuations: piano rolls with velocity, CC1, and pedal, plus shared-library renderings and each model's codec audio. Demo page: `https://temporary-brisk-iron-dzj8a7y.vercel.app/`.

5. AI USAGE STATEMENT

AI tools assisted with language drafting and LaTeX preparation. The authors specified the research design and verified the technical content, claims, and references.

6. REFERENCES

[1] Q. Lyu, B. Cruz, J. Wagner, and C.-E. Cella, "Evaluating tokenization strategies for expressive classical piano performance generation," in *Proceedings of the 29th International Conference on Digital Audio Effects (DAFx-26)*, Cambridge, MA, USA, 2026.

[2] C. Raffel, *Learning-based methods for comparing sequences, with applications to audio-to-midi alignment and matching*. Columbia University, 2016.

[3] C. Hawthorne, A. Stasyuk, A. Roberts, I. Simon, C.-Z. A. Huang, S. Dieleman, E. Elsen, J. Engel, and D. Eck, "Enabling factorized piano music modeling and generation with the maestro dataset," in *Proc. Int. Conf. Learning Representations*, 2019.

[4] Q. Kong, B. Li, J. Chen, and Y. Wang, "Giantmidi-piano: A large-scale midi dataset for classical piano music," *Transactions of the International Society for Music Information Retrieval*, vol. 5, no. 1, pp. 87–98, 2022.

[5] Y. Li, R. Yuan, G. Zhang, Y. Ma, X. Chen, H. Yin, C. Xiao, C. Lin, A. Ragni, E. Benetos *et al.*, "Mert: Acoustic music understanding model with large-scale self-supervised training," in *International Conference on Learning Representations*, vol. 2024, 2024, pp. 12 181–12 204.

[6] R. Castellon, C. Donahue, and P. Liang, "Codified audio language modeling learns useful representations for music information retrieval," in *Proc. Int. Society for Music Information Retrieval Conf.*, 2021.

[7] D. Yang, J. Tian, X. Tan, R. Huang, S. Liu, H. Guo, X. Chang, J. Shi, S. Zhao, J. Bian, Z. Zhao, X. Wu, and H. Meng, "UniAudio: Towards universal audio generation with large language models," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 56 422–56 447.

[8] A. Défossez, J. Copet, G. Synnaeve, and Y. Adi, "High fidelity neural audio compression," *Transactions on Machine Learning Research*, 2023.

[9] J. Copet, F. Kreuk, I. Gat, T. Remez, D. Kant, G. Synnaeve, Y. Adi, and A. Défossez, "Simple and controllable music generation," *Advances in neural information processing systems*, vol. 36, pp. 47 704–47 720, 2023.

[10] D. Bogdanov, M. Won, P. Tovstogan, A. Porter, and X. Serra, "The MTG-Jamendo dataset for automatic music tagging," in *Machine Learning for Music Discovery Workshop, ICML*, 2019.

[11] J. Thickstun, Z. Harchaoui, and S. Kakade, "Learning features of music from scratch," *arXiv preprint arXiv:1611.09827*, 2016.

[12] J. Ryu, S. Rhyu, H.-G. Yoon, E. Kim, J. Y. Yang, and T. Kim, "Mid-fild: Midi dataset for fine-level dynamics," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 1, 2024, pp. 222–230.

[13] B. Li, X. Liu, K. Dinesh, Z. Duan, and G. Sharma, "Creating a multitrack classical music performance dataset for multimodal music analysis: Challenges, insights, and applications," *IEEE Transactions on Multimedia*, vol. 21, no. 2, pp. 522–535, 2018.